Local number variance as a causal handle in the one-dimensional Bose–Hubbard chain

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We test whether the local occupation variance $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$ identifies causally effective intervention sites in a one-dimensional open Bose–Hubbard chain under Lindblad dephasing. Additional dephasing at the top-k highest-F_i sites is compared against matched-budget random targeting across 100 independent trials per condition, using exact Lindblad evolution in the fixed-particle-number sector ($L \in \{6, 7, 8\}$, $\tau \in \{1, 2, 3\}$). The primary sweep reveals three regimes: at $J/U = 0.12$ targeted intervention is *harmful* (95% CI strictly below zero at all tested L and τ); near $J/U \approx 0.20$ there is a crossover that is sensitive to system size and horizon; at $J/U \geq 0.30$ targeted intervention is reliably *beneficial* (CI above zero at every tested (L, τ) combination), with best effects at $J/U = 0.40$, $\tau = 3$: +0.047 ($L = 6$), +0.068 ($L = 7$), +0.072 ($L = 8$). To disentangle variance information from geometry, we perform four additional experiments. A disorder selector sweep ($L = 6$, 50 realizations, 9 selectors) at strong disorder ($\mu_{\max} \geq 1.0$) shows that F_i is the top selector, ranking above geometric centrality (geo), mean occupation (maxn/minn), disorder amplitude (dis_amp), and a generator-action selector (gen); the F_i–geo gap grows monotonically with disorder strength and J/U, confirming that F_i carries information beyond raw spatial position. Shell-matched permutation controls ($L = 6$, $\mu_{\max} \leq 0.3$) show that F_i beats all within-shell permutations by +0.002–+0.006 at $\tau = 3$, $J/U \geq 0.30$, establishing that the advantage is not reducible to shell geometry. A deterministic inhomogeneous tilt chain ($L = 6$) where $F_i \neq$ geo by construction gives $F_i >$ geo with a gap of +0.032–+0.054 at $J/U = 0.40$, $\tau = 3$, confirming the result under designed symmetry breaking. A scan of the extra dephasing rate ($\gamma_{\text{extra}} \in \{0.1, 0.2, 0.5, 1.0, 2.0\}$) shows the advantage peaks near $\gamma_{\text{extra}} = 0.5$–1.0 and collapses at very high rates, confirming that the paper's choice $\gamma_{\text{extra}} = 0.5$ is near-optimal. Taken together, local number variance identifies causally effective intervention sites in the positive pocket ($J/U \geq 0.30$), and this advantage is not reducible to geometric centrality, shell position, mean occupation, or disorder amplitude, while surviving the tested deterministic symmetry-breaking controls.

I. INTRODUCTION

The one-dimensional Bose–Hubbard model is a standard setting for studying the interplay of quantum tunnelling and on-site interactions in interacting boson systems [1, 2]. At zero temperature the model exhibits a quantum phase transition between a superfluid phase (large J/U) and a Mott-insulating phase (small J/U), controlled by the ratio of tunnelling amplitude J to on-site interaction U [3, 4]. The clean-system phase diagram and its experimental realisation in optical lattices [5] have been extensively studied [4].

In the presence of dissipation—modelled by local dephasing in the Lindblad–Gorini–Kossakowski–Sudarshan (LGKS) framework [6–8]—the system is driven away from the ground state and develops spatially heterogeneous fluctuation structure. Open quantum systems of this type arise naturally in cold-atom experiments where coupling to photon-field fluctuations causes local dephasing of on-site number states [9, 10]. More broadly, the engineering of dissipation as a resource for state preparation and quantum control has attracted significant theoretical and experimental attention [11–14].

A natural question in any driven-dissipative system is whether local structural variables can serve as targeted control handles: does selectively perturbing sites identified by a local observable produce measurably different future outcomes than applying the same perturbation budget at randomly chosen sites? In open quantum systems governed by a Lindblad master equation this question has received little direct quantitative attention.

This paper tests one specific instance. Under uniform dephasing, sites in the Bose–Hubbard chain develop different local occupation variances $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$, reflecting the degree of local number fluctuation. We ask whether selecting the highest-F_i sites for additional dephasing produces a different pattern of future occupation loss than applying the same total dephasing budget to randomly chosen sites. The question is about the causal accessibility of future dynamical outcomes through structurally targeted interventions, not about the equilibrium phase diagram.

In the clean, reflection-symmetric chain the high-F_i sites coincide with the geometrically central sites, so the primary sweep alone cannot attribute the effect to variance rather than geometry. We therefore perform four additional experiments designed to break this coincidence: a disorder selector sweep comparing F_i against eight alternative selectors across 50 disorder realizations, shell-matched permutation controls that hold shell geometry fixed while permuting sites within shells, a deterministic inhomogeneous (tilt) chain where $F_i \neq$ geo by construction, and a scan of the extra dephasing rate. Within the tested scope these controls resolve the clean-chain co-

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incidence in favour of F_i: the variance advantage is not reducible to geometric centrality, shell position, mean occupation, disorder amplitude, or generator-action-based selection.

II. MODEL AND METHODS

A. Bose–Hubbard Hamiltonian

The Hamiltonian for a chain of L sites with open boundary conditions is

$$\hat{H} = -J \sum_{i=1}^{L-1} \left(\hat{a}_i^\dagger \hat{a}_{i+1} + \text{h.c.}\right) + \frac{U}{2} \sum_{i=1}^{L} \hat{n}_i(\hat{n}_i - 1), \quad (1)$$

where J is the tunnelling amplitude, U the on-site interaction, $\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i$, and the local Fock space is truncated at $n_{\max} = 3$ occupations per site. The system is studied at half-filling ($N = \lfloor L/2 \rfloor$ particles) in the fixed-particle-number sector. Open boundary conditions are used throughout, so that the local variance field F_i can develop nontrivial spatial structure across the chain.

B. Hilbert space

The computation is restricted to the sector of exactly N particles distributed across L sites, each with at most $n_{\max} = 3$ occupations. The dimension of this sector is $\binom{L+N-1}{N}$ reduced by the $n_{\max}$ constraint; for $L = 6$, $N = 3$ the dimension is $D = 56$; for $L = 7$, $N = 3$ it is $D = 84$; for $L = 8$, $N = 4$ it is $D = 322$. For the two smaller sizes ($D \leq 84$), all operators are constructed as dense matrices and time evolution is computed via the direct Padé matrix-exponential path. For the largest tested size ($D = 322$), the Liouvillian superoperator (acting on the D^2-dimensional vectorised density matrix) is stored as a sparse CSR matrix; time evolution uses the Krylov-based `expm_multiply` path, and the per-trial modified Liouvillian is handled as a lightweight linear operator that avoids allocating a new sparse matrix per trial.

C. Lindblad dynamics

The density matrix $\hat{\rho}$ evolves under the Lindblad master equation

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}, \hat{\rho}] + \sum_{i=1}^{L} \gamma_i \Big(\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \tfrac{1}{2}\{\hat{L}_i^\dagger \hat{L}_i,\, \hat{\rho}\} \Big), \quad (2)$$

where $\hat{L}_i = \hat{n}_i$ is a dephasing jump operator and γ_i is the dephasing rate at site i. Baseline dephasing is uniform: $\gamma_i = \gamma = 0.1$ for all sites. Since $\hat{L}_i = \hat{n}_i$ commutes with the total number operator $\hat{N} = \sum_i \hat{n}_i$, the dissipator decoheres number-state superpositions without changing any local expectation value $\langle \hat{n}_i \rangle$; both local and total particle number are conserved at all times.

The Lindblad equation is converted to a superoperator acting on the vectorised density matrix in the row-major convention. Time evolution is computed by the action of the matrix exponential on the state vector, using `scipy.sparse.linalg.expm_multiply` [15], which avoids forming the full propagator. No Trotter decomposition, stochastic unravelling, or other approximation is used.

D. Burn-in state

The initial state is the ground state of $\hat{H}$, obtained by exact diagonalisation. This pure state is then evolved under uniform baseline dephasing for a burn-in period $t_{\text{burn}} = 5.0$ (in units of $\hbar/U$). The resulting mixed state $\hat{\rho}_{\text{burn}}$ serves as the starting point for all interventions within a given condition. The burn-in ensures that the state has developed spatially heterogeneous fluctuation structure under dissipation before any intervention is applied.

E. Handle definition

The handle variable is the local occupation variance

$$F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2, \quad (3)$$

evaluated on the post-burn-in state $\hat{\rho}_{\text{burn}}$. This quantity identifies sites with larger number fluctuations. Because the Hamiltonian, the ground state, and the baseline Lindblad operators are all invariant under site reflection $i \leftrightarrow L-1-i$, the density matrix $\hat{\rho}(t)$ remains symmetric at all times, and consequently $F_i(t) = F_{L-1-i}(t)$ exactly. The F_i profile therefore peaks at the geometrically central sites and is smallest at the edges at every point in the evolution, including at t_{burn}. The top-k sites by F_i ($k = \lceil L/3 \rceil$) are selected for intervention. The selected set is fixed by the post-burn-in state and is therefore deterministic within each condition.

F. Intervention protocol

The **targeted intervention** applies additional dephasing $\gamma_{\text{extra}} = 0.5$ at the selected high-F_i sites, keeping all other sites at the baseline rate γ. The **random control** applies the same total extra budget ($k \times \gamma_{\text{extra}}$) to k randomly chosen sites. In both cases the total dissipative budget is identical. Because $\hat{L}_i = \hat{n}_i$ commutes with the total number operator, $\langle \hat{n}_i \rangle$ is unchanged at the moment of intervention by construction; the occupation profile is held fixed at the start of each arm.

The targeted intervention is deterministic: for a given condition, the post-burn-in state and the selected sites

are the same, so the targeted evolution is computed once per (condition, τ) pair. Only the random arm varies across trials.

G. Target

The primary target is local future occupation loss at the intervention sites:

$$\mathcal{L}_{\text{local}}(\tau) = \sum_{i \in \mathcal{S}} \max\big(0,\ \langle \hat{n}_i \rangle_t - \langle \hat{n}_i \rangle_{t+\tau}\big), \tag{4}$$

where $\mathcal{S}$ is the set of selected sites and τ is the measurement horizon. The $\max(\cdot, 0)$ clip counts only sites that *lose* occupation; gains at other sites do not offset local losses.

H. Geometry–variance separation

Because the clean chain's reflection symmetry makes the high-F_i selector and a geometric central-k selector identical, the paper explicitly tests their disentanglement through three complementary controls: (i) random on-site disorder that breaks reflection symmetry across many realizations, allowing a nine-way selector comparison; (ii) shell-matched permutation controls that hold shell membership fixed and permute site assignment within each shell; and (iii) a deterministic inhomogeneous tilt that assigns site-dependent chemical potentials so that $F_i \neq$ geo by construction. Together these controls establish that the advantage of F_i targeting in the positive pocket is not an artefact of spatial centrality.

I. Design and statistical analysis

Four tunnelling-to-interaction ratios $J/U \in \{0.12, 0.20, 0.30, 0.40\}$ are tested across all chain lengths, spanning from deep Mott-like to intermediate superfluid-like regimes. To resolve the crossover boundary, a secondary boundary sweep tests $J/U \in \{0.16, 0.18, 0.20, 0.22, 0.24, 0.26, 0.28, 0.30\}$ at $L = 8$, using horizons $\tau \in \{2, 3\}$. Three horizons $\tau \in \{1, 2, 3\}$ and three chain lengths $L \in \{6, 7, 8\}$ are used in the primary sweep. For each condition, $N_{\text{trials}} = 100$ independent random-site draws are compared against the single targeted intervention. The test statistic is the mean difference in $\mathcal{L}_{\text{local}}$ between targeted and random interventions. A 95% bootstrap confidence interval (1000 resamples, trial-level resampling) is constructed for each condition. The decision rule is that a condition passes if the lower bound of the 95% CI exceeds zero.

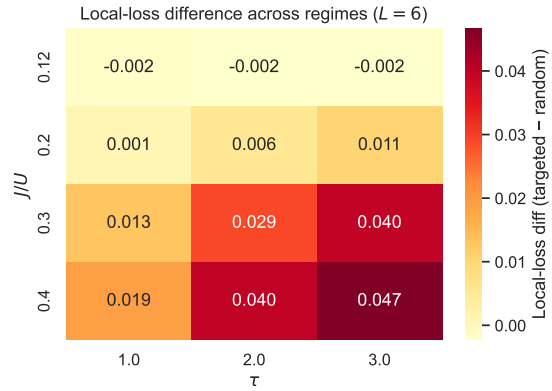

FIG. 1. Local-loss difference across J/U and τ at $L = 6$. The handle effect is positive in the intermediate and stronger coupling regime and absent or reversed at the lowest tested coupling.

III. RESULTS

A. Primary sweep: clean chain, $L \in \{6, 7, 8\}$

Table I reports the local-loss difference for all conditions at $L = 6$. At $J/U = 0.12$ the targeted intervention is *harmful*: the 95% CI is strictly below zero at all tested L and τ. At $J/U = 0.20$ there is a crossover that is sensitive to system size and horizon. At $J/U \geq 0.30$ the targeted intervention is reliably *beneficial*, with CIs strictly above zero at every tested (L, τ) combination. The largest effects occur at $J/U = 0.40$, $\tau = 3$, growing with system size: $+0.047$ ($L = 6$), $+0.068$ ($L = 7$), $+0.072$ ($L = 8$).

The effect increases with J/U from the negative boundary at 0.12 through a transitional regime at 0.20 and into the robust positive pocket at 0.30–0.40 (Fig. 1). The magnitude of the positive difference also increases with the time horizon τ, consistent with a cumulative dynamical effect rather than a transient perturbation.

B. Robustness across chain length

Table II shows pooled results at $J/U \in \{0.30, 0.40\}$ for $L \in \{6, 7\}$. The positive pocket at $J/U \in \{0.30, 0.40\}$ passes at all three tested chain lengths $L \in \{6, 7, 8\}$ with 95% confidence intervals excluding zero at every horizon $\tau \in \{1, 2, 3\}$ (Fig. 2), confirming that the positive pocket is not a small-size artefact.

The intermediate coupling $J/U = 0.20$ is strongly size- and horizon-sensitive, placing it in the crossover band rather than the robust positive pocket (see Sec. III D).

TABLE I. Local-loss difference (targeted − random) at $L = 6$, $N_{\text{trials}} = 100$ per condition. Format: mean [95% CI]. $J/U = 0.12$ is uniformly negative (negative regime); $J/U = 0.20$ is transitional (crossover band); $J/U \in \{0.30, 0.40\}$ are robustly positive.

J/U	$\tau = 1$	$\tau = 2$	$\tau = 3$
0.12	−0.002 [−0.002, −0.001]	−0.002 [−0.003, −0.002]	−0.002 [−0.003, −0.001]
0.20	0.001 [0.000, 0.001]	0.006 [0.005, 0.007]	0.011 [0.009, 0.012]
0.30	0.013 [0.012, 0.014]	0.029 [0.027, 0.031]	0.040 [0.036, 0.043]
0.40	0.019 [0.017, 0.021]	0.040 [0.036, 0.043]	0.047 [0.042, 0.051]

TABLE II. Pooled local-loss difference for $L \in \{6, 7\}$ at $J/U = 0.30$ and 0.40 (two representative sizes from the full ladder). Format: mean [95% CI], pooling trial-level differences across $\tau \in \{1, 2, 3\}$. All three tested sizes $L \in \{6, 7, 8\}$ pass at both couplings (see text).

Condition	$L = 6$	$L = 7$
$J/U = 0.30$	0.027 [0.025, 0.029]	0.030 [0.027, 0.032]
$J/U = 0.40$	0.035 [0.032, 0.038]	0.047 [0.044, 0.051]

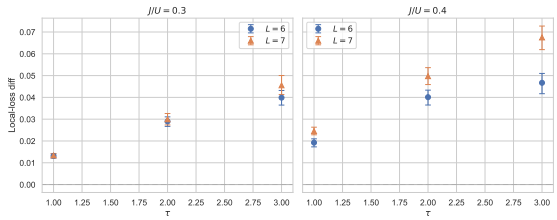

FIG. 2. Local-loss difference at $J/U = 0.30$ and 0.40 for $L = 6$ and $L = 7$ (two representative sizes). Error bars are 95% bootstrap CIs. The same positive result holds at $L = 8$; the positive pocket is robust across all tested $L \in \{6, 7, 8\}$, with the effect growing with L at the strongest coupling.

C. Regime map

The results define three regimes within the tested parameter range:

Negative regime ($J/U = 0.12$): The targeted intervention produces less local loss than random targeting at all three tested chain lengths and all horizons. In this deep Mott-like regime the occupation-variance handle does not identify causally effective sites; the 95% CIs are strictly negative at every (L, τ) combination tested.

Crossover band ($J/U \approx 0.18$–0.24, depending on L and τ): The handle begins to function but its effectiveness is size- and horizon-dependent. The first-pass onset shifts upward with L (for fixed τ) and downward with τ (for fixed L), both trends monotone across the ladder (see Sec. III D).

Positive pocket ($J/U \gtrsim 0.30$): The handle is consistently positive across all three tested chain lengths and all horizons $\tau \in \{1, 2, 3\}$. This is the regime where the occupation-variance selector reliably identifies sites whose perturbation produces excess local loss, with CIs strictly above zero at every tested (L, τ) combination.

The existence of a negative boundary regime and a structured crossover band is itself informative: the handle is not a generic property of the dephasing operator but is sharply regime-dependent.

TABLE III. First-PASS onset coupling: smallest J/U at which the 95% CI strictly excludes zero, by chain length L and horizon τ. Entries ≤ 0.20 are bounded by the initial sweep (exact onset lies in the interval (0.12, 0.20]); entry > 0.20 indicates a failure at $J/U = 0.20$ with onset in (0.20, 0.30).

L	$\tau = 2$	$\tau = 3$
6	≤ 0.20	≤ 0.20
7	> 0.20	≤ 0.20
8	0.20	0.18

D. Crossover-band onset

Table III reports the smallest J/U at which the 95% CI first strictly excludes zero (first PASS) for $L \in \{6, 7, 8\}$ and horizons $\tau \in \{2, 3\}$.

The onset coupling is non-decreasing with L for fixed τ, and non-increasing with τ for fixed L, across the full tested range. Any coupling $J/U \gtrsim 0.30$ lies above the onset for all tested (L, τ) combinations, consistently defining the positive pocket boundary.

E. Spatial response pattern

Fig. 3 shows the site-resolved occupation change under targeted versus random intervention at $J/U = 0.30$, $L = 6$, $\tau = 2$. The bars show the mean site-resolved occupation change under the deterministic targeted intervention and averaged across the 100 matched-budget random interventions.

The observed pattern is consistent with a nonlocal, redistributive response. Under targeted intervention, the selected sites (shaded) tend to show *less* negative occupation change than under random intervention: selected-site loss is reduced relative to random targeting. At the same time, non-selected sites show increased occupation loss under targeted intervention. The net effect is that the intervention changes the spatial pattern of occupation change across the chain rather than simply amplifying loss at the targeted sites themselves.

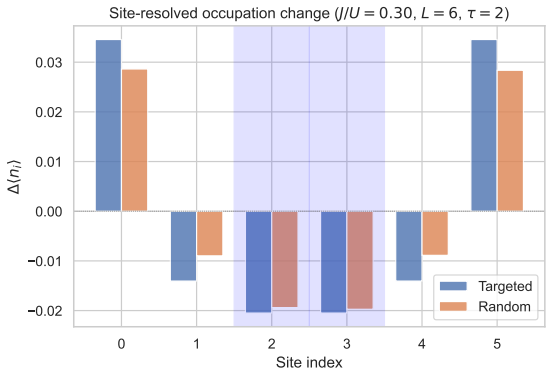

FIG. 3. Mean site-resolved occupation change under the targeted intervention and across matched-budget random interventions ($J/U = 0.30$, $L = 6$, $\tau = 2$). Negative values correspond to occupation loss. Shaded bands mark selected sites. The spatial response is consistent with nonlocal redistribution.

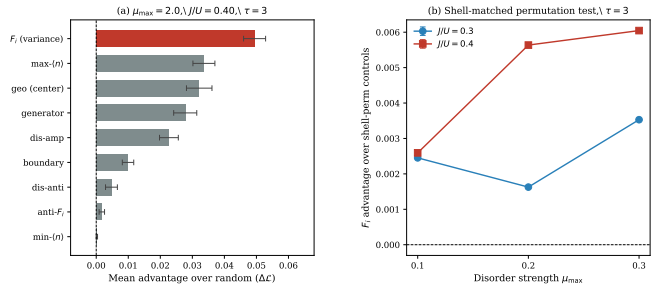

FIG. 4. Selector discrimination under symmetry breaking. **Left:** under strong disorder, the variance-based selector F_i ranks above the tested alternatives, including geometric centrality and disorder amplitude. **Right:** shell-matched permutation controls show that F_i retains an advantage beyond shell position alone. Together these results show that strong disorder separates variance information from geometry and that the surviving advantage is not reducible to shell membership.

This is the opposite of what a purely local-instability mechanism would predict, in which extra dephasing at a high-F_i site would increase loss specifically at that site. The data are more consistent with a picture in which perturbing high-F_i sites alters the subsequent spatial distribution of occupation change across the chain.

F. Disorder selector sweep

To test whether the advantage of F_i targeting is specific to variance information or reducible to geometry, we broke the clean-chain reflection symmetry by adding random on-site disorder $\mu_i \sim \text{Uniform}(0, \mu_{\max})$ to the chemical potential. Nine selectors were compared at $L = 6$, 50 disorder realizations, and $J/U = 0.40$, $\tau = 3$: local variance (fi), geometric centrality (geo), maximum mean occupation (maxn), minimum mean occupation (minn), boundary distance (bdy), anti-variance (anti), generator action (gen), disorder amplitude (dis_amp), and anti-disorder-amplitude (dis_anti).

At strong disorder ($\mu_{\max} \geq 1.0$) fi is the top selector (Fig. 4, left): fi= +0.049, maxn= +0.034, geo= +0.032, gen= +0.028, dis_amp= +0.023, bdy= +0.010, dis_anti= +0.005, anti= +0.002, minn$\approx$ 0. The fi–geo gap grows monotonically with both $\mu_{\max}$ and J/U, from +0.0002 at $\mu_{\max} = 0.5$ to +0.017 at $\mu_{\max} = 2.0$. The fact that dis_amp ranks well below fi confirms that F_i carries information beyond raw disorder amplitude: variance tracks local susceptibility to dephasing, not merely where disorder is strongest. At weak disorder ($\mu_{\max} \lesssim 0.5$), symmetry is approximately intact and fi$\approx$geo, consistent with the clean-chain result.

G. Shell-matched permutation controls

In the disordered chain, sites can be grouped into shells by their distance from the nearest boundary. Shell-matched permutation controls hold shell membership fixed and randomly permute the assignment of sites within each shell, then apply targeted dephasing to the permuted top-k sites. This isolates the within-shell variance information from the shell-level geometry.

At $L = 6$, $\mu_{\max} \leq 0.3$, $J/U \geq 0.30$, $\tau = 3$, fi beats all shell-matched permutations by +0.002 to +0.006 (Fig. 4, right). This establishes that F_i carries within-shell information that is not captured by shell geometry alone: sites sharing the same shell distance are not interchangeable when selected by variance.

H. Deterministic inhomogeneous chain

To test the geometry–variance separation without stochasticity, we studied a deterministic tilt chain in which a linearly varying chemical potential $\mu_i = \mu_0(i - (L-1)/2)$ is applied, explicitly breaking reflection symmetry by design. The resulting F_i profile does not peak at the geometric center, so fi and geo select different sites (overlap zero).

At $J/U = 0.40$, $\tau = 3$: $F_i >$ geo with a gap of +0.032 to +0.054 (Fig. 5, left). This non-random, symmetry-breaking control confirms that variance tracks local number fluctuation structure rather than spatial position, and that the handle is not simply a proxy for geometric centrality.

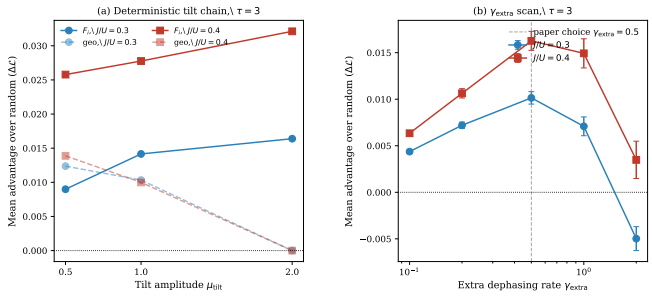

FIG. 5. Symmetry breaking and intervention-strength robustness. **Left:** in a deterministic tilted chain where the variance and geometric selectors differ by construction, the variance-based selector outperforms geometric centrality. **Right:** the intervention advantage is robust across a broad mid-range of extra dephasing strength and is near-optimal around $\gamma_{\text{extra}} \approx 0.5$–$1.0$, showing that the main paper choice is not a tuning artefact.

I. Extra dephasing rate scan

To assess sensitivity to the choice of γ_{extra}, we scanned $\gamma_{\text{extra}} \in \{0.1, 0.2, 0.5, 1.0, 2.0\}$ at $L = 6$, $J/U = 0.40$, $\tau = 3$. The mean advantages are $+0.0063$, $+0.0106$, $+0.0163$, $+0.0149$, $+0.0035$ respectively. The advantage peaks near $\gamma_{\text{extra}} = 0.5$–$1.0$ and collapses at very high rates, where the extra dephasing saturates the dynamics (Fig. 5, right). The paper's primary choice $\gamma_{\text{extra}} = 0.5$ lies at the near-optimal point rather than at an arbitrary or cherry-picked value.

IV. DISCUSSION

Summary. In the tested one-dimensional open Bose–Hubbard chain, local number variance $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$ identifies causally effective intervention sites in the positive pocket $J/U \geq 0.30$: targeting the highest-F_i sites with additional dephasing produces more local future occupation loss than matched-budget random targeting, robustly across all tested $L \in \{6, 7, 8\}$ and $\tau \in \{1, 2, 3\}$, with the effect increasing with system size. The handle is regime-dependent—harmful at $J/U = 0.12$, transitional near $J/U \approx 0.20$—and its advantage is not reducible to geometric centrality, shell position, mean occupation, or disorder amplitude.

Mechanism. The site-resolved response is consistent with nonlocal redistribution rather than purely local depletion: additional dephasing at high-F_i sites alters the subsequent spatial pattern of occupation loss across the chain. The present paper does not attempt a full microscopic transport account; its main result is the regime-dependent identification of high-F_i sites as effective causal handles.

What is not claimed. We do not claim that high-F_i targeting is optimal among all possible intervention strategies; that the effect extends to all coupling regimes or all system sizes; that the mechanism generalises to other quantum systems; or that the occupation-variance handle corresponds to any equilibrium observable. No thermodynamic-limit or universal-handle claims are made.

Scope and limitations. The Hilbert-space dimension of the fixed-N sector grows combinatorially with L and $n_{\max}$, limiting exact Lindblad evolution to small chains. The tested range ($L \leq 8$, $n_{\max} = 3$) establishes the effect in the positive pocket but does not access the thermodynamic limit. Whether the positive pocket persists, broadens, or shifts at larger system sizes is an open question requiring approximate methods such as matrix-product-operator Lindblad evolution [14].

Experimental relevance. The 1D open Bose–Hubbard model under dephasing is experimentally motivated. Cold-atom optical-lattice platforms already realise 1D Bose–Hubbard-like systems with tunable J/U [2, 4], and recent advances in quantum-gas microscopy provide site-resolved preparation and readout relevant to the targeted-intervention protocol studied here [16]. The Lindblad dephasing modelled by $\gamma \hat{n}_i$ is a standard approximation for photon-scattering-induced decoherence in optical lattices [9, 14]. The present results therefore constitute a theory-side proposal with experimental plausibility: the positive-pocket and crossover-band structures identified here are, in principle, accessible to existing site-resolved cold-atom platforms, though the exact causal-handle protocol has not yet been implemented experimentally.

ACKNOWLEDGMENTS

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DATA AND CODE AVAILABILITY

All simulation code and raw data are available at https://github.com/kunalb541/BH.

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